

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I Sandhiya.K /192521378 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	1	1	1	1
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

24/30

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Scenario

Smart campus Network (192.168.100.0/24)

Given:

A smart campus has been assigned the IP address block 192.168.100/24. The administrator has allocated the following subnet masks:

- * School of computing $\rightarrow /25$
- * School of Electronics $\rightarrow /26$
- * School of Engineering $\rightarrow /27$

Solution:

1. School of computing (/25)

* Network Address: 192.168.100.0

* Broadcast Address: 192.168.100.127

* Valid Host IP range: 192.168.100.1 to 192.168.100.126

* Usable Host Address: 126.

2. School of Electronics (/26)

* Network Address: 192.168.100.128

* Broadcast Address: 192.168.100.191.

* Valid Host IP range: 192.168.100.193 to 192.168.100.222

* Usable Host Address: 62.

3. School of Mechanical Engineering (127)

* Network Address: 192.168.100.192

* Broadcast Address: 192.168.100.223

* Valid Host IP range: 192.168.100.193 to 192.168.100.222

* Usable host Addresses: 30

4. Remaining unused IP Address Range

After allocating the above subnets the unused IP address range is:

192.168.100.224 to 192.168.100.255

This remaining range can be

reserved for future departments or network expansion.

Multinational software company (10.10.0.0/16)

Given:

- * Software Development $\rightarrow /24$
- * Quality Assurance $\rightarrow /25$
- * Human Resource $\rightarrow /26$
- * Executive Management $\rightarrow /24$

1. Software Development (/24)

The /24 subnet

- * Network Address: 10.10.0.0
- * Broadcast Address: 10.10.255
- * Valid Host IP Range: 10.10.0.1 to 10.10.0.254
- * Usable Hosts: 254

2. Quality Assurance (/25)

- * Network Address: 10.10.0.0
- * Broadcast Address: 10.10.127
- * Valid Host IP Range: 10.10.0.1 to 10.10.0.126
- * Usable Hosts: 126

Smart Hospital IPv6 Migration

Given:

2001:0db8:abc:0000:0000:0000:0000/64.

1. compressed IPv6 Address

compressed Address:

2001:db8:abcd::/64

2. Expand IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000

3. Network Prefix

with a prefix length of /64, the first 64 bits represent the network portion.

Network Prefix: 2001:db8:abcd:0::/64.

4. Host portion

The remaining 64 bits are used for host addressing.

Solution:

The destination IP address 192.168.2.45 belongs to the subnet 192.168.2.0/24 because it falls within the range 192.168.2.1 to 192.168.2.254.

The router checks its routing table and finds the matching entry 192.168.2.0/24.

The packet is then forwarded through the router interface connected to the 192.168.2.0/24 network.

The destination IP address does not match any routing table entry, the route (0.0.0.0/0).

are: suitable computer IP addresses

PC1: 192.168.10.2

PC2: 192.168.10.3

PC3: 192.168.10.4

The default gateway for all computers is 192.168.10.1.

Finance LAN:

The /26 Subnet also provides

62 usable host addresses

* Network Address: 192.168.10.64

* Broadcast Address: 192.168.10.127

* Valid Host IP Range: 192.168.10.65 to

192.168.10.126.

Three suitable computer IP

addresses are:

* PC1: 192.168.10.66

* PC2: 192.168.10.67

* PC3: 192.168.10.68

The default gateway for all

finance computer is

192.168.10.65.